Practical Work No. 4: Backup and Restore in SQL Server

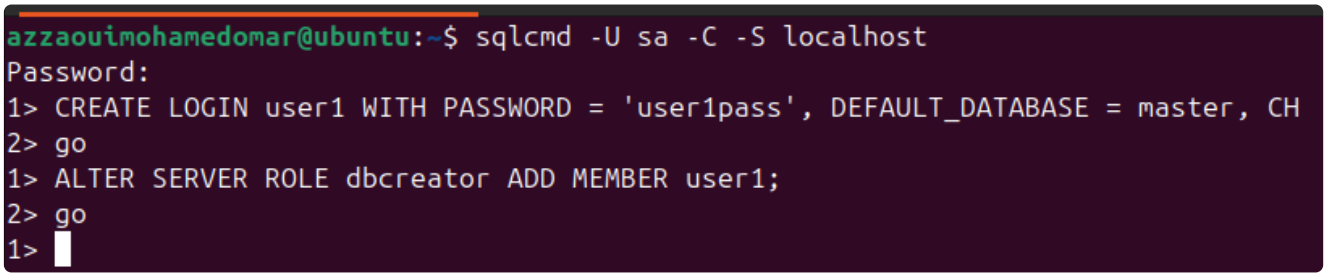
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Course: Database Administration
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Exercise 1: Full backup and log backup with a logical device

For this exercise we worked on SQL Server 2025 running on Ubuntu, using sqlcmd and DBeaver as clients.

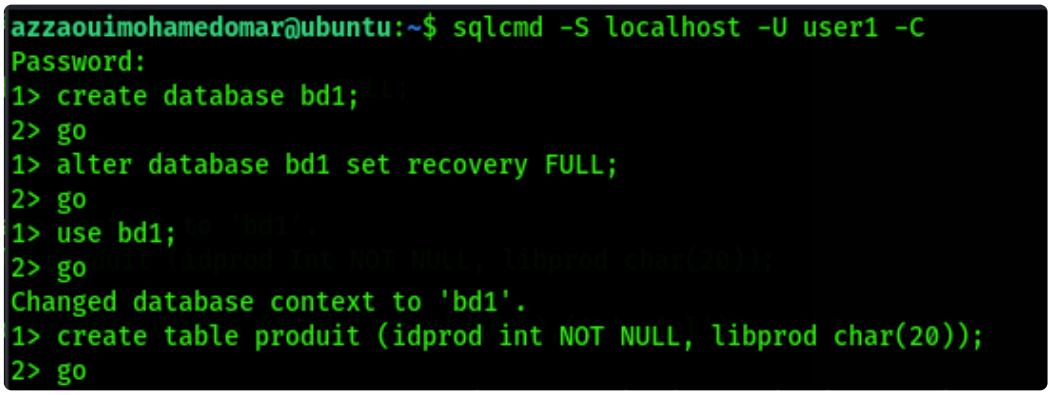
1. Creating the login user1

As sa (super admin), we created the login user1 and gave it the server role dbcreator, which allows it to create databases.



2. Creating the database BD1 and the table produit

Connected as user1, we create the database bd1 and switched it to the FULL recovery model. This choice matters because only the FULL model allows transaction log backups, which we need later to restore the database up to the last saved moment. I then created the table `produit` and inserted three records.



```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U user1 -C
Password:
1> use bd1;
2> go
Changed database context to 'bd1'.
1> INSERT INTO produit VALUES (1, 'ariel'), (2, 'megaphone'), (3, 'telephone');
2> go

(3 rows affected)
```

3. Creating the login user2 and giving it its roles

As sa, we create the login user2 and give it the server roles dbcreator, needed to restore databases, and diskadmin, needed to manage backup devices.

```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U sa -C
Password:
1>  WITH PASSWORD = 'user2pass', DEFAULT_DATABASE = master, CHECK_POLICY = OFF;
2> go
1> alter server role dbcreator ADD MEMBER user2;
2> go
1> alter SERVER role diskadmin ADD MEMBER user;
2> go
Msg 156, Level 15, State 1, Server ubuntu, Line 1
Incorrect syntax near the keyword 'user'.
1> alter SERVER role diskadmin ADD MEMBER user2;
2> go
```

Then, inside bd1, we declared user2 as a database user and added it to the role db_backupoperator, which gives it the right to back up this database.

```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U sa -C
Password:
1> USE bd1;
2> go
Changed database context to 'bd1'.
1> create user user2 for login user2;
2> go
1> alter role db_backupoperator add member user2;
2> go
1>
```

4. Creating the logical backup device Backup1

Connected as user2, we create a logical backup device named Backup1 with sp_addumpdevice. A logical device is simply a name that points to a physical file, so later we can only write TO Backup1 instead of the full path. we then check that it was indeed created by querying the view sys.backup_devices.

```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U user2 -C
Password:
1> exec sp_addumpdevice 'disk', 'Backup1', '/var/opt/mssql/tests/SauverBD.bak';
2> go
1> select name,physical_name From sys.backup_devices;
2> go
name
                                                physical_name
------------------------------------------------------------------------------------------------------------------------------------
------------------------------------ ------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------------------------------------------------------------------------
----------------------------------------
Backup1
                                                /var/opt/mssql/tests/SauverBD.bak
```

5. Full backup of BD1

Still as user2, we made a full backup of bd1 on the device Backup1.

```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U user2 -C
Password:
1> backup database bd1 TO Backup1 WITH init;
2> go
Processed 416 pages for database 'bd1', file 'bd1' on file 1.
Processed 1 pages for database 'bd1', file 'bd1_log' on file 1.
BACKUP DATABASE successfully processed 417 pages in 0.201 seconds (16.176 MB/sec).
1>
```

6. Adding more records

As user1, we added more products. These rows were inserted after the full backup, so they are not contained in it. Only the log backup will be able to bring them back.

```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U user1 -C
Password:
1> INSERT INTO bd1.dbo.produit VALUES (4, 'plush'), (5, 'spoon'), (6, 'martir');
2> go

(3 rows affected)
1>
```

7. Backing up the transaction log

As user2, we back up the transaction log directly into the file BD1_LOG1.trn.

```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U user2 -C
Password:
1> backup log bd1 to disk = '/var/opt/mssql/tests/BD1_LOG1.trn';
2> go
Processed 1 pages for database 'bd1', file 'bd1_log' on file 1.
BACKUP LOG successfully processed 1 pages in 0.022 seconds (0.310 MB/sec).
1>
```

Finally, we list the backup folder to confirm that both files were really written to disk.

```
azzaouimohamedomar@ubuntu:~$ sudo ls -l /var/opt/mssql/tests/
[sudo] password for azzaouimohamedomar:
total 3504
-rw-rw---- 1 mssql mssql   84480 Oct 11 15:46 BD1_LOG1.trn
-rw-rw---- 1 mssql mssql 3496448 Oct 11 15:43 SauverBD.bak
```

8. Simulating a disk failure

As sa, we put bd1 offline. Then, in the Ubuntu terminal, we deleted the two files of the database, BD1.mdf and BD1_log.ldf, to simulate a disk crash. From now on the database can only be brought back from our backups.

```
root@ubuntu:/home/azzaouimohamedomar# sudo ls -l /var/opt/mssql/data/ | grep "bd1*"
-rw-rw---- 1 mssql mssql  8388608 Oct 11 15:48 bd1_log.ldf
-rw-rw---- 1 mssql mssql  8388608 Oct 11 15:48 bd1.mdf
-rw-rw---- 1 mssql mssql 14680064 Oct 12 09:20 model_msdbdata.mdf
-rw-rw---- 1 mssql mssql 16187392 Oct 12 09:28 msdbdata.mdf
root@ubuntu:/home/azzaouimohamedomar# /var/opt/mssql/data/
bash: /var/opt/mssql/data/: Is a directory
root@ubuntu:/home/azzaouimohamedomar# cd /var/opt/mssql/data/
root@ubuntu:/var/opt/mssql/data# rm bd1_log.ldf bd1.mdf
root@ubuntu:/var/opt/mssql/data# ls -l /var/opt/mssql/data/ | grep "bd1*"
-rw-rw---- 1 mssql mssql 14680064 Oct 12 09:20 model_msdbdata.mdf
-rw-rw---- 1 mssql mssql 16187392 Oct 12 09:28 msdbdata.mdf
```

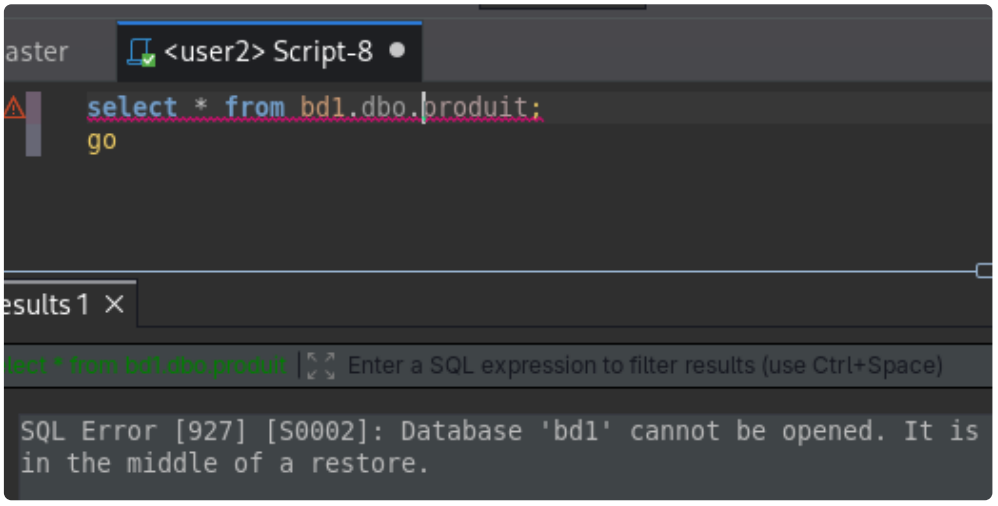
```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U sa -C
Password:
1> alter database BD1 SET offline with ROLLBACK immediate;
2> go
```

9. Restoring the full backup with NORECOVERY

As user2, we restored the full backup of bd1 from the device Backup1 with the option NORECOVERY. This option tells SQL Server that more backups are coming, so the restore is not finished yet.

```
azzaouimohamedomar@ubuntu:~$ sqlcmd -S localhost -U user2 -C
Password:
1> restore database bd1 from Backup1 with NORECOVERY,replace;
2> go
Processed 416 pages for database 'bd1', file 'bd1' on file 1.
Processed 1 pages for database 'bd1', file 'bd1_log' on file 1.
RESTORE DATABASE successfully processed 417 pages in 0.050 seconds (65.029 MB/sec).
1> select name,state_desc from sys.databases where name = 'bd1';
2> go
name                                                             state_desc
---------------------------------------------------------------- ----------
bd1                                                              RESTORING

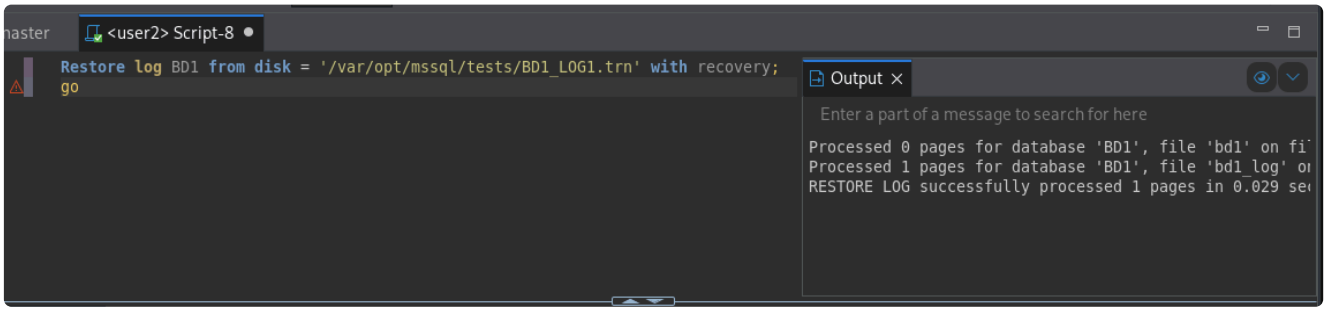
(1 rows affected)
```

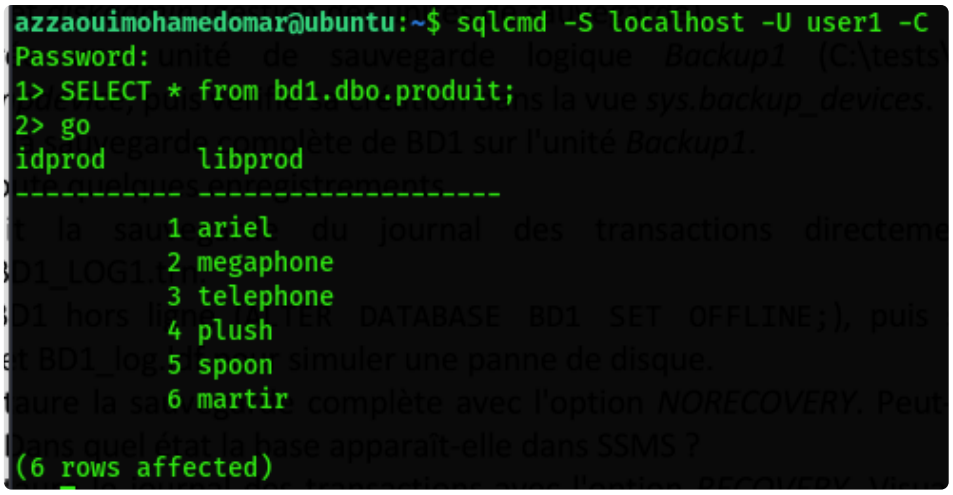


Answer: you can't query the table. In DBeaver the database shows that it's in the middle of a restore.

10. Restoring the transaction log with RECOVERY

As user2, we restored the log backup BD1_LOG1.trn with the option RECOVERY. This time the restore is finished, so SQL Server rolls back any unfinished transaction and brings the database back online. We then displayed the table produit.





The table contains all the products again, including the ones added after the full backup (products 4 to 6). They were recovered thanks to the log backup, which shows that the full backup and the log backup together bring the database back to its last saved state.